FACULTYHIRE SYSTEM DOSSIER

Comprehensive Technical Architecture, Polyglot Engineering, and Defense Report

SYSTEM	ARCHITECTURE	CORE STACK	VERIFICATION
FacultyHire v2.4	Polyglot Microservices	Java 17 + Python 3.10	22/22 Automated Tests

1. Executive Summary & Problem Space

Higher education faculty recruitment differs fundamentally from corporate industrial hiring. Standard Applicant Tracking

? Terminal Degree Misclassification: Interdisciplinary or allied doctorates (e.g. Applied Math & Computational Science applying

? Neglect of the Academic Triad: Faculty candidates must be evaluated across Teaching & Pedagogy, Research &

Systems (ATS) rely heavily on exact keyword matching, arbitrary resume scoring, and rigid years-of-experience checklists.

? Absence of Committee Tools: Search committees need actionable probing interview questions, ABET accreditation

for Software Engineering) are flagged as 'Ph.D. missing' by naive keyword filters.

? Air-Gapped Privacy Constraints: Universities frequently manage confidential dossiers under strict privacy regulations

Publications, and Curriculum Stewardship, rather than an arbitrary single percentage.

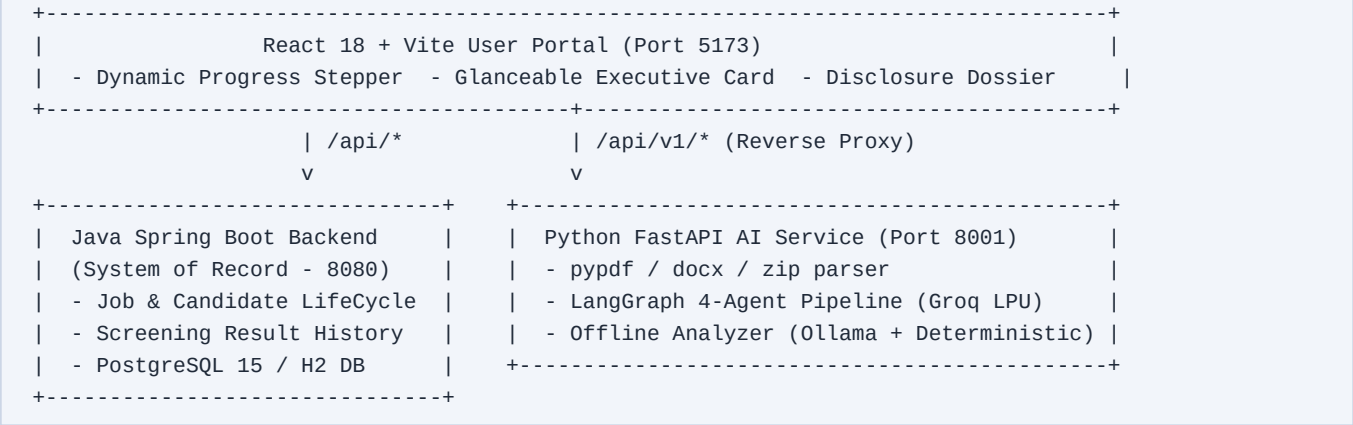
[CASE STUDY AUDIT] Real-World Case Study: Dr. Marcus Vance

requiring 100% on-premise, air-gapped operation without cloud leaks.

2. High-Level System Architecture

FacultyHire adheres to the System of Record (SoR) architectural pattern. It decouples long-term transactional data persistence

Figure 1: Decoupled Polyglot Microservices Topology



3. Core Subsystems Deep Dive

3.1 Component A: Java Spring Boot 3 Core Backend (Port 8080)

Developed with Java 17 and Spring Boot 3.x, this service enforces business invariants, transactional safety, and data integrity.

? Dual-Profile Persistence: Production profile connects to PostgreSQL (recruitment_db). Air-gapped fallback profile activates an

? Domain Models: Includes Job, Candidate, Resume, and ScreeningResult JPA entities with strict relational constraints.

3.2 Component B: Python AI Cognitive Microservice (Port 8001)

It exposes RESTful endpoints for job requisitions, candidate profiles, and screening result archives.
Built with FastAPI, Pydantic v2, and LangGraph. Houses the document extraction engine and dual screening execution modes:

embedded H2 in-memory database with zero installation.

? Online Mode (Groq LPU + LangGraph): Chains 4 autonomous agents: Profile Extractor -> Rubric Analyzer -> Evidence

? Offline Air-Gapped Mode: Interfaces locally with Ollama (LLaMA 3.2 on port 11434). Falls back instantly (<5ms) to a

3.3 Component C: React + Vite Web Portal (Port 5173)

A zero-emoji, high-density web portal engineered to avoid recruiter fatigue through progressive disclosure:

? Multi-Phase Stepper: Live progress bar tracking: Upload & Parsing -> Criteria Matching -> Synthesis -> Finalizing Dossier.
Evaluator -> Scoring & Synthesis. Delivers sub-1.5s inference.

deterministic academic recruiter engine when offline.

4. The Academic Recruiter Evaluation Framework

To evaluate faculty dossiers rigorously, the engine implements two quantitative models designed for academic institutions:

4.1 4-Tier Education Crediting Matrix

Tier	Academic Attainment	Credit	Committee Action
Tier 1	Direct Discipline Ph.D. (CS, SE)	100%	Full qualification met without reservation
Tier 2	Allied STEM Doctorate (Math, EE, Physics)	75%	Research & quantitative credit; discipline ga
Tier 3	Terminal Master's (M.S. in CS/SE)	60%	Teaching track credit; lacks research termina
Tier 4	Related Bachelor's Degree only	30%	Lacks prerequisites for tenure-track faculty

4.2 Three-Pillar Academic Scoring Formula

The overall score is calculated as a calibrated weighted sum reflecting the tripartite mission of university faculty:

Equation 1: Academic Triad Composite Formulation

```
Composite Academic Match Score = (0.40 * S_teaching) + (0.35 * S_research) + (0.25 * S_domain)

Where:
  S_teaching : Pedagogy, university lectures, course delivery, and ABET accreditation experience
  S_research : Peer-reviewed scholarship (IEEE, ACM, SIAM), terminal degree rigor, and grants
  S_domain   : Alignment with core department syllabus (e.g. Distributed Systems, Architecture)
```

4.3 Search Committee Probing Interview Generation

Every evaluation produces three tailored, probing questions addressing candidate portfolio gaps, interdisciplinary research

5. Technology Stack & Dependencies Catalog

Subsystem	Technologies	License	Architectural Role
Frontend UI	React 19, Vite 8.3, CSS3	MIT	Responsive SPA, proxying, dossier drawer
AI Microservice	FastAPI, Pydantic v2, Uvicorn	BSD	REST API, schema validation, document parser
LLM Orchestration	LangGraph, LangChain-Groq	MIT	Stateful multi-agent DAG execution
Cloud Inference	Groq LPU (Llama 3.3 / GPT-OSS)	Commercial	Sub-1.5s low-latency JSON structured outputs
Offline LLM	Ollama Daemon (LLaMA 3.2)	MIT	100% on-premise air-gapped local inference
Core Backend	Java 17, Spring Boot 3, JPA	Apache 2.0	Transactional System of Record and CRUD
Persistence	PostgreSQL 15+, H2 Database	Open Source	Relational persistence & zero-config fallback

6. End-to-End Screening Data Flow

Screening Lifecycle Sequence

```
1. Recruiter initiates single resume (PDF/DOCX) or bulk ZIP upload via React UI.
2. React displays the 4-phase animated progress stepper and sends multipart POST to /api/v1/screen-resume.
3. Vite reverse proxy routes request to Python FastAPI on Port 8001.
4. Document Parser extracts text using pypdf or python-docx with whitespace preservation.
5. Execution Router checks ANALYZER_BACKEND:
   - Online: LangGraph multi-agent pipeline coordinates Groq cloud inference.
   - Offline: Local Ollama daemon or deterministic academic rule engine runs.
6. Validated Pydantic ScreeningResult returned to UI with Three-Pillars and 3 probing questions.
7. React UI renders Glanceable Card and Dossier Drawer; saves result to Spring Boot (Port 8080) for audit.
```

7. Presentation & Oral Defense Guide

Use this structured script when explaining the project to examiners, faculty review panels, or recruiters:

? 1-Minute Pitch: 'Most recruitment platforms treat university professors like corporate programmers, discarding candidates

? Why Java + Python?: 'Enterprise systems require distinct tools for distinct workloads. Java Spring Boot provides robust

over keyword mismatches. FacultyHire was engineered for university search committees. It scores candidates across Teaching,

? Zero Hallucination Safeguards: Our architecture uses two defenses: Agent 3 is required to extract verbatim resume citations

? Air-Gapped Operation: 'When disconnected from the internet or running on private campus networks, the system operates

transactional data persistence and audit trails (System of Record). Python delivers cutting-edge NLP, document parsing, and

8. Operations & Test Suite Verification

Research, and Curriculum readiness, honors allied quantitative doctorates, and equips committees with probing interview

for every matched criterion. Furthermore, LLM inference runs with greedy decoding (temperature 0.0) and rigid JSON schemas,

FacultyHire provides turn-key Windows launcher batch scripts for both production and offline modes:

? start_all.bat: Automated launcher: Clears port conflicts, starts AI Service (8001), Spring Boot Backend (8080), and React

completely offline via start_all_offline.bat. It boots an embedded H2 database and runs local Ollama or our deterministic engine

LangGraph multi-agent orchestration. Decoupling them into microservices ensures independent scalability.'

packs.'

preventing speculative outputs.'

Frontend (5173).

in <50ms.'